User Story : Database Management

As a system administrator, I want to schedule and monitor comprehensive automatic database backups with multiple recovery options so that I can ensure complete data protection, maintain system reliability, and guarantee business continuity in case of any system failures or disasters .

Acceptance Criteria

Acceptance Criterion 1: Automated Daily Backup Execution

- Given I have configured automatic backups to run daily at 3:00 AM during low-usage hours
- When the system clock reaches the scheduled backup time
- Then the system automatically initiates a full database backup, verifies backup integrity, and logs the completion status with timestamp and file size details

Acceptance Criterion 2: Multiple Backup Types and Retention

- Given the backup system is configured for comprehensive data protection
- When daily backups are executed
- Then the system creates full backups daily, incremental backups every 6 hours, maintains 30 days of daily backups, 12 weeks of weekly backups, and 12 months of monthly archives
- Acceptance Criterion 3: Backup Failure Detection and Alerting
- Given an automatic backup process is running
- When a backup fails due to disk space, network issues, or database locks
- Then the system immediately sends critical alerts to administrators via email and SMS, logs detailed error information, and attempts automatic retry after 30 minutes

Acceptance Criterion 4: Backup Verification and Integrity Testing

- Given a backup has been completed successfully
- When the backup process finishes
- Then the system automatically performs integrity checks, validates backup file completeness, and runs monthly restore tests to verify backup usability

Acceptance Criterion 5: Secure Offsite Storage Management

- Given backups contain sensitive healthcare data requiring HIPAA compliance
- When backups are created

- Then the system encrypts all backup files, transfers copies to secure offsite storage, and maintains encrypted local and cloud-based copies with access logging

Acceptance Criterion 6: Disaster Recovery Planning Integration

- Given the hospital requires rapid recovery capabilities
- When I access the disaster recovery module
- Then the system provides automated recovery procedures, estimated recovery times, step-by-step restoration guides, and allows point-in-time recovery options

Acceptance Criterion 7: Backup Performance Monitoring and Reporting

- Given I need to monitor backup system performance
- When I access backup management reports
- Then the system displays backup success rates, storage utilization trends, backup duration analytics, and generates monthly compliance reports for healthcare regulations